Circuit Lab, Division B

Practice Test 6 (State/National level) · written test · 50 minutes · 50 questions

Team name: _____ Team number: _____

Names: _____

Instructions

- Write all answers on the answer sheet at the back. Only the answer sheet is graded.
- Give numeric answers in metric (SI) units with the correct number of significant figures.
- You may use two calculators and your own paper notes.
- Use $k = 8.99 \times 10^{9}\ \text{N·m}^2/\text{C}^2$ and $e = 1.602 \times 10^{-19}\ \text{C}$. Wires and meters are ideal unless the question says otherwise.
- In logic questions, A′ means NOT A, AB means A AND B, and A + B means A OR B.

Section	Questions
Section I: Multiple choice	1–15
Section II: Short answer and calculations	16–37
Section III: Circuit problems	38–50

Section I: Multiple choice

Circle the one best answer. 1 point each.

1. In 1820 André-Marie Ampère showed that two long parallel wires carrying currents in the SAME direction:

   A. Repel each other
   B. Have no force between them
   C. Attract each other
   D. Twist until they are at right angles

2. As a student in Berlin, Heinrich Hertz studied under which two famous physicists?

   A. Hermann von Helmholtz and Gustav Kirchhoff
   B. Michael Faraday and Humphry Davy
   C. Georg Ohm and André-Marie Ampère
   D. Alessandro Volta and Luigi Galvani

3. Georg Simon Ohm learned mathematics and physics at home from his father. What was his father's trade?

   A. Shoemaker
   B. Locksmith
   C. Silk merchant
   D. Military engineer

4. Of the seven scientists named in the rules (Ampère, Coulomb, Kirchhoff, Volta, Ohm, Hertz, Boole), which two have NO SI unit named after them?

   A. Hertz and Boole
   B. Ampère and Kirchhoff
   C. Ohm and Volta
   D. Kirchhoff and Boole

5. A technician who repairs computer boards wears a wrist strap joined to ground through a large resistor. What is the main reason?

   A. Charge on her body drains away slowly, so no spark (ESD) jumps into the chips
   B. It stops her from getting a 120 V shock
   C. It makes her body a better insulator
   D. It adds charge to the chips so they work better

6. An electroscope is already charged NEGATIVE, so its leaves are spread. A charged rod is brought near the knob (it never touches). The leaves spread FARTHER apart. The rod is:

   A. Positive
   B. Negative
   C. Neutral
   D. It cannot be told from this

7. Why is touching a 120 V wire with wet hands much more dangerous than with dry hands?

   A. Water raises the voltage of the wire
   B. Wet skin changes AC into DC
   C. Wet skin has a much lower resistance, so a much larger current flows
   D. Water stores charge like a battery

8. Which of these is a PRIMARY cell (made to be used once, not recharged)?

   A. Alkaline AA cell
   B. NiMH AA cell
   C. Lithium-ion phone battery
   D. Lead-acid car battery

9. One joule (1 J) is the same as:

A. 1 W / s
B. 1 W · s
C. 1 V / A
D. 1 A · s

10. Three equal resistors are in series with an ideal battery. A wire is then connected across ONE of them. The battery current:

A. Becomes 1.5 times as large
B. Becomes 3 times as large
C. Becomes 2/3 as large
D. Does not change

11. A refrigerator light uses a normally-closed (NC) momentary pushbutton. Which is true?

A. Its contacts conduct only while it is pressed
B. It stays in its new position after you let go
C. Its contacts conduct when it is NOT pressed, and open while it is pressed
D. It has two separate circuits (two poles)

12. A small glass fuse is marked "5 A 250 V". What does the "250 V" mean?

A. The fuse blows when the voltage passes 250 V
B. The highest voltage at which the fuse can safely break the circuit (stop the arc) when it blows
C. The fuse drops 250 V when it carries 5 A
D. The fuse can only be used on 250 V circuits

13. What mainly sets the color of the light that an LED gives off?

A. The semiconductor material (its band gap)
B. The size of the series resistor
C. The color of the plastic lens only
D. The length of its legs

14. A conducting silicon diode carries 10 mA at 0.70 V. The current is raised to 20 mA. The diode voltage becomes about:

A. 0.35 V
B. 0.70 V exactly
C. 0.72 V
D. 1.40 V

15. A 14-pin 7408 chip contains:

A. Four 2-input NAND gates
B. Six NOT gates
C. Four 2-input OR gates
D. Four 2-input AND gates

Section II: Short answer and calculations

Write the answer with units and the correct number of significant figures. 2 points each.

16. $q_1 = +4.0\ \mu C$ is at the origin. $q_2 = -6.0\ \mu C$ is at (0.30 m, 0) and $q_3 = +5.0\ \mu C$ is at (0, −0.20 m). Find the size of the net force on q_1, and give its x and y parts.

17. A +6.0 nC charge is at x = 0 and a −3.0 nC charge is at x = 0.30 m. Find the electric field (size and direction) at x = 0.10 m.

18. Two identical small metal spheres carry +6.0 μC (sphere A) and −10.0 μC (sphere B). They touch and are separated. How many electrons moved, and in which direction?

19. A car battery has an EMF of 12.7 V and an internal resistance of 0.0150 Ω. The starter motor draws 180 A. Find the battery's terminal voltage while the motor cranks.

20. A 12.0 V, 7.0 Ah lead-acid battery runs a 15 W camp light. To protect the battery, only 50% of its capacity may be used. How many hours can the light run?

21. US household AC has a frequency of 60.0 Hz. (a) What is its period? (b) How many times each second is the voltage exactly zero?

22. A US 120 V outlet has three wires. For each one give its usual insulation color and its job. Which wire normally carries no current?

23. An aluminum bus bar ($\rho = 2.65 \times 10^{-8}$ Ω·m) is 2.00 m long. Its cross-section is a rectangle 5.0 mm by 20.0 mm. Find its resistance.

24. An electric car has a 64.0 kWh battery. A home charger draws 7.20 kW from the wall and takes it from 20% to 80% full. Charging is 90.0% efficient. Electricity costs $0.150 per kWh. Find the charging time and the cost.

25. (a) A 4-band resistor reads violet, green, orange, silver. Give its value and the range it may have. (b) Two SMD resistors are marked R47 and 1R0. Give their values.

26. A meter movement reads full scale at 100 μA and has a resistance of 1.50 kΩ. What resistor, in series with it, makes a 0 to 10.0 V voltmeter?

27. Four points are joined in every possible way by six 10.0 Ω resistors (a square with both diagonals; the diagonals do not touch where they cross). Find the resistance between two neighboring corners.

28. A person touches the hot wire of a 120 V circuit while standing on a damp floor. The path through the person to ground has 15.0 kΩ. Find the current. Will a GFCI on this circuit trip?

29. A long hallway light must be switched on and off from FIVE places. How many 3-way switches and how many 4-way switches are needed?

30. A zener regulator: a 14.00 V supply, a 270 Ω series resistor, and a 6.20 V zener diode across the load. What is the SMALLEST load resistance for which the zener still holds the load at 6.20 V?

31. Use a K-map to find the simplest sum of products for F(A, B, C, D) = Σm(1, 2, 3, 7, 10, 11, 14, 15) with a don't care d(6).

32. Add the binary numbers 11010110 + 01111011. Give the sum in binary and in decimal.

33. F(A, B, C) = Σm(0, 3, 4, 5) is built with a 4-to-1 multiplexer. A and B are the select inputs (A is the high bit). What must each data input I_0 to I_3 be (0, 1, C or C′)?

34. A warning light F turns on when EXACTLY ONE of three sensors A, B, C reads 1. Write F as the simplest sum of products.

35. A power bank holds 10 000 mAh at 3.70 V. It charges a phone battery of 3200 mAh at 3.85 V. Only 85% of the bank's energy reaches the phone battery. How many FULL charges can it give the phone (from empty)?

36. A 9.0 V battery, a switch and three equal resistors R1, R2, R3 are in series. The switch is closed but no current flows. A voltmeter reads 0 V across R1, 9.0 V across R2 and 0 V across R3. What is wrong?

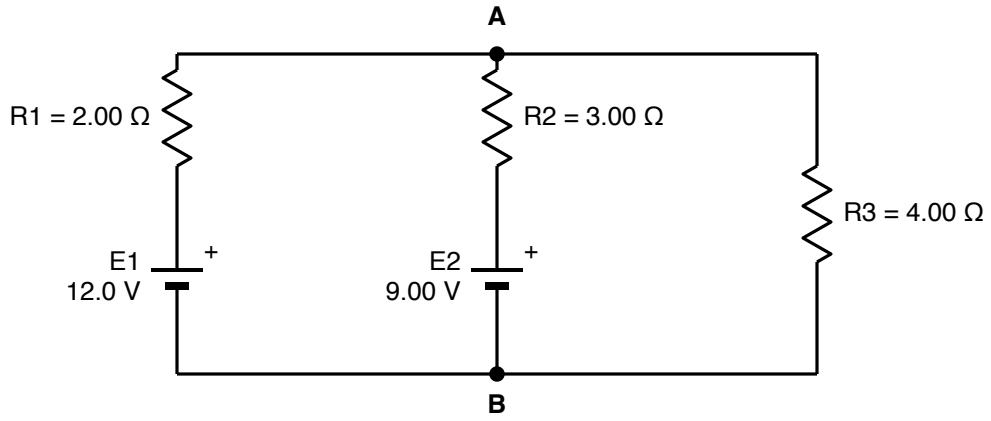
37. You have one each of 47 Ω, 100 Ω, 330 Ω and 680 Ω. Using EXACTLY THREE of them, make a resistance within 1% of 120 Ω. Show the arrangement and its exact value.

Section III: Circuit problems

Show your work. 2 points per part.

Problem A: two batteries in parallel branches (Kirchhoff's laws) (Questions 38–41)

Three branches join node A (top) and node B (bottom). Left: R1 = 2.00 Ω and E1 = 12.0 V. Middle: R2 = 3.00 Ω and E2 = 9.00 V. Right: R3 = 4.00 Ω. Both batteries have their + plate at the TOP. Let I_1 and I_2 be the currents flowing UP through E1 and E2.



38. Write two loop equations in I_1 and I_2 and find I_1.

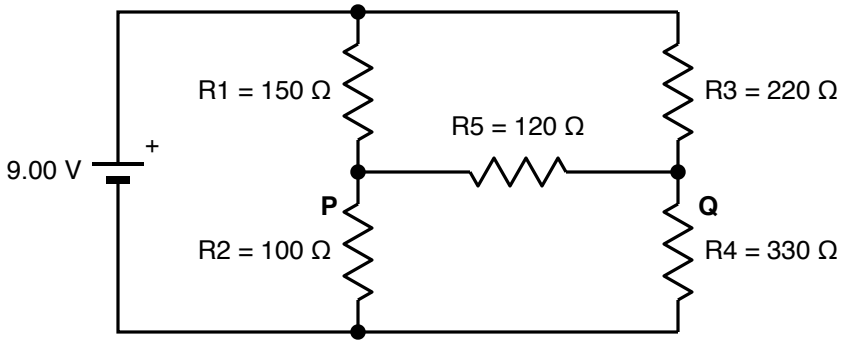
39. Find I_2. Is E2 giving power to the circuit or being charged?

40. Find the voltage of A relative to B and the current in R3.

41. How much power does E1 deliver?

Problem B: an unbalanced bridge (Questions 42–45)

A 9.00 V ideal battery feeds the bridge below. P is between R1 = 150 Ω (top) and R2 = 100 Ω (bottom). Q is between R3 = 220 Ω (top) and R4 = 330 Ω (bottom). R5 = 120 Ω joins P and Q.



42. First, R5 is removed. Find $V_P - V_Q$. Which point is higher?

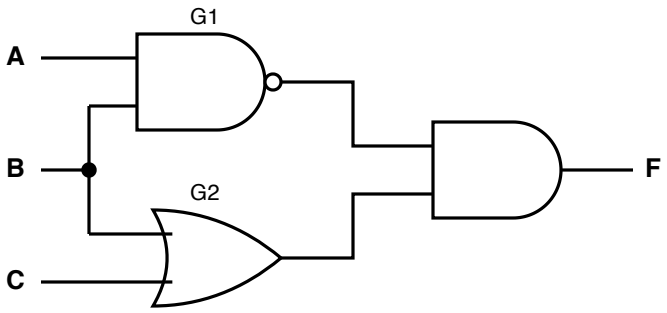
43. Now R5 is in place. Write the node equations for P and Q, and find the current in R5 and its direction.

44. With R5 in place, find the equivalent resistance that the battery sees.

45. To what value must R4 be changed so that R5 carries no current?

Problem C: a logic circuit (Questions 46–48)

G1 = NAND(A, B), G2 = OR(B, C), and F = AND(G1, G2).



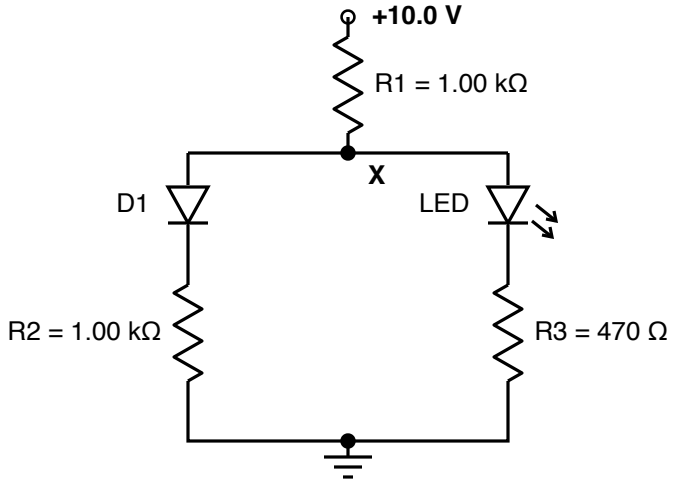
46. Write F and simplify it to the shortest sum of products.

47. Give F as a list of minterms, Σm( ).

48. F can be made by one 2-to-1 multiplexer with B as its select input. What goes to data input I_0 (used when B = 0) and to I_1 (used when B = 1)?

Problem D: a diode network (Questions 49–50)

+10.0 V feeds R1 = 1.00 kΩ to node X. From X to ground: branch 1 is a silicon diode D1 (0.70 V when on) in series with R2 = 1.00 kΩ; branch 2 is a red LED (2.00 V when on) in series with R3 = 470 Ω.



49. Assume both D1 and the LED are on. Find V_X, and check that the assumption is right.

50. Find the current in the LED.

End of the test. Check your answer sheet.

Answer sheet · Practice Test 6 (State/National level)

Team name: _____ Team number: _____

Multiple choice: circle one letter. Other questions: write the answer with units.

1.	**A**	**B**	**C**	**D**	**26.**	_____
2.	**A**	**B**	**C**	**D**	**27.**	_____
3.	**A**	**B**	**C**	**D**	**28.**	_____
4.	**A**	**B**	**C**	**D**	**29.**	_____
5.	**A**	**B**	**C**	**D**	**30.**	_____
6.	**A**	**B**	**C**	**D**	**31.**	_____
7.	**A**	**B**	**C**	**D**	**32.**	_____
8.	**A**	**B**	**C**	**D**	**33.**	_____
9.	**A**	**B**	**C**	**D**	**34.**	_____
10.	**A**	**B**	**C**	**D**	**35.**	_____
11.	**A**	**B**	**C**	**D**	**36.**	_____
12.	**A**	**B**	**C**	**D**	**37.**	_____
13.	**A**	**B**	**C**	**D**	**38.**	_____
14.	**A**	**B**	**C**	**D**	**39.**	_____
15.	**A**	**B**	**C**	**D**	**40.**	_____
16.	_____				**41.**	_____
17.	_____				**42.**	_____
18.	_____				**43.**	_____
19.	_____				**44.**	_____
20.	_____				**45.**	_____
21.	_____				**46.**	_____
22.	_____				**47.**	_____
23.	_____				**48.**	_____
24.	_____				**49.**	_____
25.	_____				**50.**	_____